Review of vortices in **wildland fire**

JM Forthofer, SL Goodrick - Journal of Combustion, 2011 - Wiley Online Library

… fire **behavior** and safety concerns. In this paper, the current state of knowledge of the interaction of **wildland fire** … In this study a computer **model** of the atmospheric boundary layer was …

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Wildfire likelihood's elements: a literature **review**

M Mhawej, G Faour, J Adjizian-Gerard - Challenges, 2015 - mdpi.com

… that affect the ignition of the **wildfire**, its initial **spread** and its free **propagation**). The first set of … generates various positive outcomes, particularly in terms of **prediction**, prevention and …

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Guidance on spatial **wildland fire** analysis: models, tools, and techniques

RD Stratton - 2006 - books.google.com

… If the **model** has not been properly calibrated to local fires, how … , when fire **behavior** outputs have been critiqued and the **model** … Definition of a fire **behavior model** evaluation protocol: a …

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Review and perspectives of digital twin systems for **wildland fire** management

Y Li, T Zhang, Y Ding, R Wadhwani… - Journal of Forestry …, 2024 - Springer

… **wildfire** prevention, monitoring, disaster response, and post-fire recovery. This **review** examines the potential utility of Digital Twin in **wildfire** … operation **model** to **model** the **wildfire spread** …

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Modeling vorticity-driven **wildfire behavior** using near-field techniques

JJ Sharples, JE Hilton - Frontiers in Mechanical Engineering, 2020 - frontiersin.org

… An example **simulation** using the pyrogenic vector potential coupled to a **wildfire spread simulation** is shown in Figure 2 with s c = 0.5 and s w = 0.5 in Equation (8). These constants …

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Wildfire autonomous response and **prediction** using cellular automata (WARP-CA)

A Ramadan - arXiv preprint arXiv:2407.02613, 2024 - arxiv.org

… realism in simulating **wildfire dynamics** [8, 16, 28]. Our literature **review** uncovered a lack of … generating heterogeneous terrain in **forest fire dynamics simulation**. The mathematical basis …

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Methods for **Wildfire Spread Prediction** and Their Integration with Remote Sensing Data

SA Khvostikov, SA Bartalev - Inf. Technol. Remote Sens. Earth …, 2019 - academia.edu

… short **review** article presents description of a field of **wildfire** … just-in-time **wildfire** danger **prediction** system. Also in last 10 years … **wildfire** models, which leads to better estimation of **wildfire** …

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Convolutional LSTM neural networks for modeling **wildland fire dynamics**

J Burge, M Bonanni, M Ihme, L Hu - arXiv preprint arXiv:2012.06679, 2020 - arxiv.org

… Here, we seek to use ML alone to explicitly **model wildland fire spread dynamics** based solely on data. In effect, ML in this context represents the development of an empirical …

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Wildfire propagation simulation tool using cellular automata and GIS

W Velasquez, A Munoz-Arcentales… - 2019 International …, 2019 - ieeexplore.ieee.org

… **Wildfire Propagation Simulation** tool able to present graphically, how the fire will be propagated in case of a **wildfire**… Calkin in ”Uncertainty and risk in **Wildland fire** management: A **review**…

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Challenges and opportunities in remote sensing-based fuel load estimation for **wildfire behavior** and management: A comprehensive **review**

A Abdollahi, M Yebra - Remote Sensing, 2025 - opus.lib.uts.edu.au

… Most fire **behavior prediction** models, including BEHAVE, FlamMap, and FARSITE require a … and **spread** of surface fires [39,46]. Dead fuels play a pivotal role in **wildfire prediction**, …

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